

Wireless security test: This test is used to identify security holes and hotspots within a Wi-Fi network.

Social engineering test: This is done to ensure that the employees of a company are following the company's training procedures and protocols, and don't fall into the trap of phishing attacks or other similar cyber threats.

Infrastructure test: This is used to check for vulnerabilities within the system as a whole.

IoT pen tests: An IoT pen test is done to protect user data globally.

PCI pen test: PCI compliance standards are a set of requirements that makes sure that all companies process, store, and transmit card information in a safe and secure environment. PCI tests help to ensure that payment data security systems meet the PCI compliance standards.

Role and importance of pen testing in security

The main objective behind pen testing is to identify issues that might eventually lead to security and data breaches. This is done to implement adequate security controls. There are also specialised testing tools available that test the robustness of the security policies of any organisation.

As it is a simulated cyber attack, it helps ethical hackers evaluate the effectiveness of the measures that an organisation has already taken. This is done to find out the vulnerabilities before the attacker does. In the case of networks, the main goal is to strengthen security. This is done by closing unused ports, troubleshooting services, and calibrating the firewall rules. This route of action helps in eliminating all types of security loopholes. In the case of a Web application, testing is used to identify, analyse and report vulnerabilities such as buffer overflow, SQL injection, cross-site scripting, etc.

Pen testing helps to test security controls and gain insights into the

overall condition of the application. It is also helpful in assessing the network and the physical security layers. Testing exposes the susceptible points in a computer system that hackers would attack in the first go. Most importantly, it helps thwart future attacks by validating essential security controls.

The stages of pen testing

Penetration testing or pen testing is used to estimate the ability of a system to protect its networks against threats. The threat could be both external and internal. There are a few stages involved in penetration testing.

Plan: The planning process starts by defining the test's precise aim and scope. To better understand the target and efficiently conduct the test, as much information as possible should be collected about the functions of the system and any potential vulnerability.

Scan: In this stage, the network is scanned using static or dynamic analysis. This step helps inform the pen testers about how a particular application responds to the various threats it encounters.

Gain access: The weaknesses of a target application are located in this stage. Various pen testing strategies like cross-site scripting and SQL injection are used in this process.

Maintain access: This stage of pen testing is used to determine the ability of cybercriminals to maintain their presence within the system after successfully breaching the security layers. This is done to see if they can gain deeper access to the system and, if yes, how long they can maintain it. This helps to assess the damage this level of attack may do, and how much data loss could happen.

Analyse: In this stage, pen testers report the outcome of the penetration test. A detailed report is prepared, containing all the exploited

vulnerabilities, all the sensitive data that has been accessed, and how much time it took for the system to respond to the attackers. Based on this report, several suggestions are given with respect to the hardware and software changes that should be made to protect the system from such attacks in the future.

Penetration testing is used quite frequently to augment the Web application firewall for Web application security. Testing for Web applications involves breaching application systems such as APIs, frontend/backend servers, and code injection attacks.

How to learn pen testing

Pen testing is essentially a form of hacking. So if you love solving puzzles and cracking codes, then pen testing is a skill you can learn. You can then legally break and hack into the computer systems of companies.

To become a professional pen tester, there are many courses available online, both free and paid. A bachelor's degree in computer science or a relevant field will also increase your chances of learning pen testing. There are certifications available for the same to help you not just advance your skills but also break into the field.

There are many different types of pen testing tools available on the Internet for free that you can use to practice on your own.

- **Vulnerability scanner:** Used to scan the environment and detect known vulnerabilities and configuration errors.
- **Web proxy:** Used as an intermediary server that separates users from the Web pages they attempt to browse.
- **Network sniffer:** Used to collect and analyse the traffic in a network.
- **Port scanner:** Used to detect open ports.
- **Password cracker:** Used to recover passwords that are stored or either transmitted in a scrambled form.

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